



Laparoscopic Instruments: A Quick Guide

A pocket reference to the kit you handled today: what each instrument looks like, what it's for, and where it appears in a laparoscopic appendicectomy and cholecystectomy.

1 · The camera (laparoscope) 5 mm or 10 mm rigid telescope



flat tip: looks straight ahead

0° scope

- Flat lens, so you see **exactly where the scope points**.
- Easiest to orientate. Good for learners, diagnostic laparoscopy and straightforward views.
- To see round a corner you have to move the whole scope.



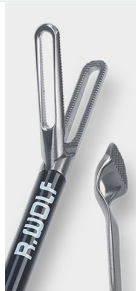
bevelled tip: view angled 30°

30° scope

- Bevelled lens, so the view is **angled 30° off the shaft**, pointing away from the light post.
- **Rotate the light lead** to look up, down or around structures, e.g. over the gallbladder into Calot's triangle, or down into the pelvis.
- Keep the **camera head upright** so the horizon stays level, and turn only the light lead.

2 · Graspers & dissectors usually 5 mm, ratcheted or non-ratcheted handles

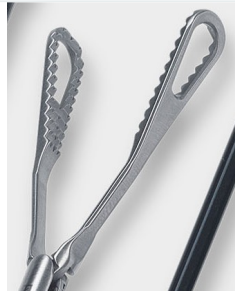
A



Atraumatic grasper **BOWEL-SAFE**

Fenestrated (Johan-type) jaws with fine, shallow teeth that spread the pressure. Use for **handling bowel** and delicate tissue, e.g. finding and lifting the appendix or holding the caecum.

B



Bull-nose grasper **BOWEL-SAFE**

Broad, blunt, rounded jaws for a gentle grip on a larger area. Use for **retracting bowel, omentum or the gallbladder** without tearing.

C



Wolf grasper **TRAUMATIC**

Toothed (crocodile-type) jaws for a firm, traumatic grip. Use **only on tissue being removed**, e.g. the gallbladder fundus or appendix tip. **Never on bowel**.

D



Maryland dissector

CAN TAKE MONOPOLAR

Curved, narrow, serrated jaws. The workhorse for **blunt dissection**: opening the window in the mesoappendix or dissecting Calot's triangle. Attach the diathermy lead to **grasp and coagulate** small bleeding points.

E

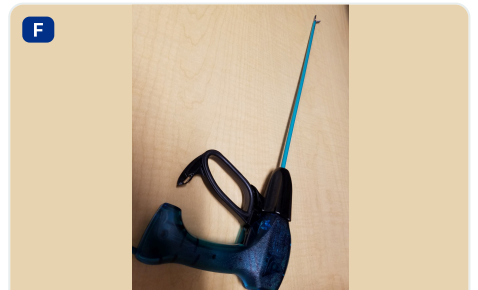


Hook diathermy (L-hook)

MONOPOLAR

An insulated rod with a bare metal hook. **Hook a thin layer of tissue, lift it away from danger, then activate**. Use it to divide peritoneum and to take the gallbladder off the liver bed.

F



Advanced energy device

BIPOLAR / ULTRASONIC

LigaSure-type (bipolar sealing) or Harmonic-type (ultrasonic) jaws that **seal and cut** vessels in one step, typically up to 5–7 mm depending on the device. Mainly used in **advanced colorectal surgery and bowel resections**.

3 · Energy Device Usage Etiquette

Monopolar (hook, or Maryland with the lead attached): current goes from the instrument tip **through the patient** to the return plate on the thigh. It's quick and versatile for dissecting and coagulating small vessels, but **it acts wherever the tip touches**.

Bipolar: current passes only **between the two jaws**, so only the tissue you grip is heated and there's less stray energy. Advanced bipolar and ultrasonic devices (LigaSure-type, Harmonic-type) seal larger vessels and are mainly used in **advanced colorectal surgery and bowel resections**.

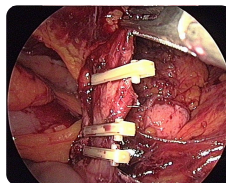
⚠ Using monopolar safely

- **Keep the active tip in view** whenever you press the pedal. Never activate off-screen.
- **Lift tissue away** from bowel and other structures before you activate.
- Don't let the active tip **touch another metal instrument or clip**, which would carry the current onward.
- Check the **instrument insulation** is intact. A crack can burn bowel outside your view.
- Use **short bursts**, in contact with tissue. Don't activate in mid-air.
- Let the hook **cool** before touching bowel, and check the **return plate** is well applied.

4 · Ports, clips & endoloops



Ports (trocars): 5 mm for most instruments; 10–12 mm for a 10 mm camera, the clip applicator and specimen retrieval. The first port is usually umbilical, placed by **open (Hasson)** technique or after a **Veress needle** pneumoperitoneum. All other ports go in **under vision**.



Clip applicator: titanium clips, or polymer locking clips (Hem-o-lok type), applied **across the whole structure** with the tips seen beyond it. In a chole: the cystic duct and cystic artery. **Endoloop**: a pre-tied slip-knot loop cinched around a tubular structure. Classically used on the **appendix base**.

Typical ports. **Appendix**: umbilical 10/12 mm (camera), suprapubic 5 mm, left iliac fossa 5 mm. **Chole**: umbilical 10/12 mm, epigastric 10 mm (or 5 mm), and two 5 mm right subcostal. Set-ups vary by surgeon.

5 · Putting it together

Laparoscopic appendicectomy

Access

Umbilical port, camera in, 2 working ports under vision

Find & lift

Atraumatic graspers along the taenia to the appendix

Window

Maryland opens the mesoappendix at the base

Mesoappendix

Hook / Maryland diathermy ± clips

Secure base

Endoloops on the base, then divide

Retrieve

In a bag through the 10/12 mm port

Laparoscopic cholecystectomy

Access

Umbilical camera port plus 3 working ports

Retract

Fundus up over the liver; Hartmann's pouch out laterally

Dissect

Hook / Maryland open the peritoneum over Calot's triangle

Critical view of safety

Only 2 structures enter the GB; lower third of the cystic plate cleared

Clip & divide

Cystic artery and duct

Off the liver

Hook diathermy along the liver bed, then retrieve in a bag

6 · Take-home points

- **Match the jaw to the tissue**: atraumatic for bowel, toothed only for what's coming out.
- **0° to orientate, 30° to look around**: turn the light lead, not the camera head.
- **Every monopolar activation should be in view**, in contact and brief.
- **No clipping or dividing in a chole** until you have the critical view of safety.